

Last name _____

First name _____

LARSON—MATH 310—CLASSROOM WORKSHEET 13
Matrix-Vector Multiplication.

Review

1. Write M as a matrix of column vectors $\begin{bmatrix} \vec{a}_1 & \vec{a}_2 \end{bmatrix}$. What are $\vec{a}_1$ and $\vec{a}_2$?
2. Write M as a matrix of row vectors vectors $\begin{bmatrix} \vec{b}_1 \\ \vec{b}_2 \\ \vec{b}_3 \end{bmatrix}$. What are $\vec{b}_1$, $\vec{b}_2$ and $\vec{b}_3$?
3. Find the diagonal matrix $\text{diag}(3, 4, 5)$.
4. Find a 3×3 *upper triangular* matrix U with all non-zero entries equal to 2.
5. Find a 3×3 *lower triangular* matrix L with all non-zero entries equal to 3.
6. Find the *transpose* M^T .
7. Find the scalar product $5M$.
8. What is the matrix $0_{3,2}$? Is it the same as $0_{2,3}$?
9. Find $\|M\|$.

New (Chp. 6)

1. What is the *definition* of $A\vec{x}$, for an $m \times n$ matrix A and a $n \times 1$ vector $\vec{x}$?

Let A be the 3×2 matrix:

$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$$

Let $\vec{x} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$.

2. Use this definition to find $A\vec{x}$.

A be the 3×2 matrix: $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$, and $\vec{x} = \begin{bmatrix} 7 \\ 8 \end{bmatrix}$.

3. Use the *row interpretation* of matrix-vector multiplication to find $A\vec{x}$.
4. Use the *column interpretation* of matrix-vector multiplication to find $A\vec{x}$.
5. Find $A\vec{e}_1$.
6. Find $A\vec{e}_2$.
7. What is I in $I\vec{x}$?
8. Find $I\vec{x}$?